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ACADEMIC DETAILS

National Institute of Technology Durgapur

Ph.D., 25 January 2021

- Thesis Title: *Multi-hop Cognitive Radio Networks with RF Energy Harvesting*
- Keywords: Cognitive Radio; Energy Harvesting; NOMA; Relay Selection; Imperfect channel estimation
- Supervisor: Prof. Sumit Kundu and Dr. Sanjay Dhar Roy
- Department: Electronics and Communication Engineering

National Institute of Technology Durgapur

M.Tech., May, 2010

- Specialization: Telecommunication Engineering, CGPA: 8.75
- Dissertation Topic: *Call Admission Control in Cellular CDMA Network*
- Keywords: Call Admission Control; Signal to Interference Noise ratio; Outage Probability
- Supervisor: Dr. Sanjay Dhar Roy
- Department: Electronics and Communication Engineering

Haldia Institute of Technology, West Bengal

B.Tech., June, 2008

- Department: Electronics and Communication Engineering, DGPA: 7.78
- University: West Bengal University of Technology (W.B.U.T)

Bidhan Chandra Institution Durgapur

Higher Secondary (Class 12), July, 2004

- Board: W.B.C.H.S.E, Percentage of Marks: 68.70%

D.T.P.S.H.S. High School, Durgapur

Madhyamik (Class 10), July, 2002

- Board: W.B.B.S.E, Percentage of Marks: 72.12%

POST-DOCTORAL RESEARCH EXPERIENCE

- ❖ **Post held:** Post-Doctoral Research Fellow in Institute of Communications Engineering, National Sun Yat-sen University, Kaohsiung, Taiwan
Employment period: March, 2024 - till date
Responsibilities: Research on "6G Communication"
- ❖ **Post held:** Institute Post-Doctoral Fellow in EEE Dept, Indian Institute of Technology Guwahati,
Employment period: December, 2021 - Dec, 2023
Responsibilities: Research on "RIS assisted Hybrid RF/Free Space Optical Communication"

PUBLICATIONS

- ❖ SCI/SCIE $\Rightarrow$ 20, ❖ SCOPUS $\Rightarrow$ 8, ❖ CONFERENCE $\Rightarrow$ 19

SCI/SCIE						SCOPUS		CONFERENCE	
IEEE			non-IEEE			IEEE	non-IEEE	IEEE	non-IEEE
Q1	Q2	Q3	Q1	Q2	Q3	0	8	19	0
7	0	0	2	9	2				

• Journals (SCI/SCIE)

- [J20.] D. Bepari, **S. Mondal**, A. Bhowal, K. Singh, H. Shin, K.k. Wong, and D. W. K. Ng “ Fluid Antennas: Pioneering a New Era in Next-Generation Wireless Communications” *IEEE Communications Surveys and Tutorials* , vol. 28, pp. 5596-5631, 2026. [SCI indexed] [IF 46.7] **Link**
- [J19.] **S. Mondal**, K. Singh, C. -P. Li and Z. Ding, “NOMA-Enhanced Active RIS-aided MISO ISAC System under NTN with Hardware,” *IEEE Transaction on Communication* , vol. 74, pp. 4629-4645, 2026. [SCI indexed] [IF 8.3] **Link**
- [J18.] Dipen Bepari , **S. Mondal**, B. Acharya, and D. Dubey , “UAV-Assisted NOMA-Enabled Relay Communication,” *International Journal of Communication Systems* , vol. 39, no. 1, pp. e70322, 2026 [SCIE indexed] [IF 1.8] **Link**
- [J17.] **S. Mondal**, K. Singh, A. Kaushik, and C.P. Li, “Performance of RIS-Aided Fluid Antenna-Enabled Multiuser NOMA Non-terrestrial Networks ,” *IEEE Journal on Selected Areas in Communications* , vol. 44, pp. 1061-1076, 2026. [SCI indexed] [IF 17.2] **Link**
- [J16.] **S. Mondal**, K. Singh, C. Pan, and C.P. Li, “Performance Analysis of STAR-IRS-aided MISO-ISAC Systems with Multiple Targets: A Rate-Splitting Approach,” *IEEE Transaction on Communication* , vol. 73, no. 10, pp. 9658-9675, Oct. 2025. [SCI indexed] [IF 8.3] **Link**
- [J15.] **S. Mondal**, D. Bepari, A. Chandra, K. Singh, C. -P. Li and Z. Ding, “ A Comprehensive Survey on NOMA-Based Backscatter Communication for IoT Applications,” *IEEE Internet of Things Journal* , vol. 12, no. 12, pp. 18929-18953, 2025. [SCI indexed] [IF 8.9] **Link**
- [J14.] **S. Mondal**, K. Singh, C. -P. Li and Z. Ding, “ Mixed FSO/Active IRS-Aided MISO NOMA Communication With Imperfect CSI and SIC,” *IEEE Internet of Things Journal* , vol. 12, no. 1, pp. 623-636, 2025. [SCI indexed] [IF 8.9] **Link**
- [J13.] D. Bepari, **S. Mondal**, A Misra, and M Guizani “ Performance analysis of CSI independent cache-aided NOMA ordering,” *Computers and Electrical Engineering* , vol. 119, pp. 109570, 2024. [SCI indexed] [IF 4.9] **Link**
- [J12.] **S. Mondal**, L Davoli, SD Roy, S Kundu, G Ferrari, R Raheli “Throughput and delay analysis of cognitive M2M communications,” *Journal of Network and Computer Applications* , vol. 225, pp. 103856, 2024. [SCI indexed] [IF 8.0] **Link**
- [J11.] D. Bepari, **S. Mondal**, P Pareek, and N Gupta “Uplink Performance Analysis of Wireless Energy Harvesting-Enabled NOMA-based Networks,” *Mobile Networks and Applications* , vol. 29, pp. 856–866, 2024. [SCIE indexed] [IF 2.4] **Link**
- [J10.] D. Bepari, **S. Mondal**, A. Chandra, R. Shukla, Y. Liu, M. Guizani, and A. Nallanathan “ A Survey on Applications of Cache-Aided NOMA,” *IEEE Communications Surveys and Tutorials* , vol. 25, no. 3, pp. 1571-1603, thirdquarter 2023. [SCI indexed] [IF 46.7] **Link**
- [J09.] N. Gupta, D. Bepari, K.Yadav. **S. Mondal** “Outage Analysis of an IRS-assisted Relaying NOMA Network,” *Iranian Journal of Science and Technology, Transactions of Electrical Engineering* , vol. 48, pp. 93–102, 2024. [SCIE indexed] [IF 1.7] **Link**
- [J08.] **S. Mondal**, S. D. Roy, and S. Kundu, “Performance of a Hybrid Dual-hop and Multi-hop Network based on NOMA,” *International Journal of Electronics*, vol. 110, no. 5, pp. 898-914, 2023. [SCIE indexed] [IF 1.1], **Link**
- [J07.] **S. Mondal**, S. D. Roy, and S. Kundu, “Partial Relay Selection in Energy Harvesting based NOMA Network with imperfect CSI,” *Wireless Personal Communications* (Springer), vol. 120, no. 4, pp. 3153—3169, 2021. [SCIE indexed] [IF 2.2], **Link**

- [J06.] **S. Mondal**, S. D. Roy, and S. Kundu, "Outage Analysis for NOMA based Energy Harvesting Relay Network with Imperfect CSI and Transmit Antenna Selection," *IET Communications* (IET Digital Library), vol. 14, no. 14, pp. 2240-2249, 2020. [SCI indexed] [IF 1.6], **Link**
- [J05.] S. Biswas, D. Bepari, **S. Mondal**, "Relay selection and performance analysis of wireless energy harvesting networks," *Wireless Personal Communications* (Springer), vol. 114, no. 4, pp. 3157-3171, 2020. [SCIE indexed] [IF 2.2], **Link**
- [J04.] S. Ghosh, **S. Mondal**, S. D. Roy, and S. Kundu, "D2D Communication with Energy Harvesting Relays for Disaster Management," *International Journal of Electronics* (Taylor & Francis), vol. 107, no. 8, pp. 1272-1290, 2020. [SCIE indexed] [IF 1.1], **Link**
- [J03.] **S. Mondal**, S. D. Roy, and S. Kundu, "Closed-Form Outage Probability Expressions for Multihop Cognitive Radio Network with Best Path Selection Schemes in RF Energy Harvesting Environment," *Wireless Personal Communications* (springer), Vol. 103, no. 3., pp. 2197-2212, 2018. [SCIE indexed] [IF 2.2], **Link**
- [J02.] **S. Mondal**, S. D. Roy, and S. Kundu, "Energy harvesting based multihop relaying in cognitive radio network," *Wireless Personal Communications* (Springer), vol. 97, no. 4, pp. 6325-6342, 2017. [SCIE indexed] [IF 2.2], **Link**
- [J01.] **S. Mondal**, S. D. Roy, and S. Kundu, "Primary behaviour-based energy harvesting multihop cognitive radio network," *IET Communications*, vol. 11, no. 16, pp. 2466-2475, 2017. [SCI indexed] [IF 1.6], **Link**
- Journals (Scopus)
- [S08.] **S. Mondal**, S. K. Choudhary, A Misra, U. Biswas, D Bepari "Outage Analysis of a Content-based User Pairing in NOMA," *International Journal of Electronics Letters* (Taylor & Francis), Vol. 13, no 3, 272-279, 2025. [Scopus indexed], **Link**
- [S07.] D Bepari, A Misra, **S. Mondal**, I Bala "Partial cooperative NOMA for improving outage performance of edge users," *International Journal of Electronics Letters* (Taylor & Francis), Vol. 12, no. 1, pp. 69-86, 2024. [Scopus indexed], **Link**
- [S06.] A Misra, I Bala, D Bepari, **S. Mondal** "A Finite HARQ-based Non-orthogonal Multiple Access for Reliable Communications," *International Journal of Electronics Letters* (Taylor & Francis), Vol. 12, no. 2, pp. 178-192, 2024. [Scopus indexed], **Link**
- [S05.] **S. Mondal**, S. D. Roy, and S. Kundu, "On performance of multihop energy harvesting CRN in presence of co-channel interferers," *International Journal of Electronics Letters* (Taylor & Francis), Vol. 10, no. 1, pp. 1-15, 2022. [Scopus indexed], **Link**
- [S04.] **S. Mondal**, S. D. Roy, and S. Kundu, " Outage and throughput performance of a multihop network with an adaptive power splitting-based energy harvesting," *International Journal of Electronics Letters* (Taylor & Francis), vol. 9, no. 4, pp. 403-412, 2021. [Scopus indexed], **Link**
- [S03.] **S. Mondal**, S. D. Roy, and S. Kundu, " Performance Analysis of a Cognitive Radio Network with Adaptive RF Energy Harvesting," *International Journal of Electronics Letters* (Taylor & Francis), Vol. 9 no. 1., pp. 114-127, 2021. [Scopus indexed], **Link**
- [S02.] **S. Mondal**, D. Bepari, and S. Biswas "Performance of a multihop cognitive radio network with diversity under imperfect CSI," *International Journal of Electronics Letters* (Taylor & Francis), vol. 11, no. 1, pp. 41-54, 2022. [Scopus indexed], **Link**
- [S01.] K. Chandra, **S. Mondal**, S. D. Roy, and S. Kundu, "Outage probability analysis of a secondary user in an underlay dual hop cognitive amplify and forward relay network," *Perspectives in Science* (Elsevier), Vol. 8, 117-120, 2016. [Scopus indexed], **Link**
- [S00.] SS Bhattacharjee, **S. Mondal**, S. D. Roy, and S. Kundu, "Comparative throughput study of energy harvesting CR network with adaptive hybrid relay," *CSI Transactions on ICT* (springer), Vol. 6, no. 2, pp.179-185, 2018., **Link**

- Conferences

- [C19.] **S. Mondal**, Keshav Singh, Aryan Kaushik and C.P. Li, "Active RIS-Aided Mixed FSO-THz NOMA Network: Performance and Statistical Analysis", Proc. of *2025 IEEE International Conference on Communications*, 24–28 May 2026, Glasgow, Scotland, UK.
- [C18.] **S. Mondal**, K. Singh, C.P. Li, and Z. Ding, "Active RIS-aided NOMA-ISAC Network Architecture for LEO Satellite Communications", Proc. of *IEEE International Conference on Information and Communication Technology Convergence (ICTC) (ICTC 2025)*, Jeju Island, Korea, Oct. 2025, pp. 1554-1559. **Link**.
- [C17.] Himanshu Tiwari, Anirban Bhowal, **S. Mondal**, Keshav Singh "Performance Analysis and Resource Optimization in Multiple RIS-Aided TeraHertz MIMO Communications", Proc. of *2025 IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 2216-2221. **Link**.
- [C16.] Shikha Maurya, **S. Mondal**, Deepak Sahu, Shailendra Singh, Matadeen Bansal, and Keshav Singh "Robust Energy-Aware Precoding and Relay Selection for MIMO-Based Shared Spectrum Networks", Proc. of *2025 IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 2302-2307. **Link**.
- [C15.] **S. Mondal**, Dipen Bepari, Shikha Maurya, Keshav Singh, and Aryan Kaushik "RIS-Assisted Transmit Antenna Selection-Spatial Modulation Network with Manifold Optimization", Proc. of *2025 IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 2197-2202. **Link**.
- [C14.] **S. Mondal**, Keshav Singh, Aryan Kaushik and Cunhua Pan "An Outage Analysis of Hovering UAV and STAR-RIS Aided NOMA ISAC System for SAGIN", Proc. of *IEEE Global Communications Conference (GLOBECOM 2025)*, Taipei, Taiwan, 2025, pp. 6460-6465. **Link**.
- [C13.] **S. Mondal**, Dipen Bepari, Keshav Singh, and Aryan Kaushik "Ergodic Capacity of Hard-Switching in Hybrid FSO/IRS-Assisted RF Transmission", Proc. of *International Symposium on Multimedia and Communications (ISMAC 2025)*, Tagbilaran, Philippines, 2025, pp. 1-6. **Link**.
- [C12.] **S. Mondal**, Keshav Singh, Derrick Wing Kwan Ng, Chih-Peng Li, and Zhiguo Ding "Outage Performance of RIS-Aided NOMA ISAC Network for LEO Satellite System", Proc. of *2025 IEEE International Conference on Communications Workshops (ICC Workshops)*, Montreal, QC, Canada, 2025, pp. 1556-1561. **Link**.
- [C11.] **S. Mondal**, K. Singh, C.P. Li, and S. Prakriya "Mixed FSO/IRS-Aided NOMA Network with Heterogeneous Channels", Proc. of *WCNC-2025 IEEE Wireless Communications and Networking Conference*, Milan, Italy, Mar, 2025, pp. 1-6. **Link**.
- [C10.] K.L. Jiang, S. Pala, **S. Mondal**, K. Singh, and C.P. Li, "Multiple Access for Spectral Efficient Active RIS-Aided ISAC Systems", Proc. of *WCNC-2025 IEEE Wireless Communications and Networking Conference*, Milan, Italy, Mar, 2025, pp. 1-6. **Link**.
- [C09.] **S. Mondal**, K. Singh, C.P. Li, and Z. Ding, "Hybrid FSO/Active IRS-aided NOMA-IoT Communications under Imperfect CSI and SIC", Proc. of *GLOBECOM - 2024 IEEE Global Communications Conference*, Cape Town, South Africa, Dec, 2024, pp. 2888-2893. **Link**.
- [C08.] **S. Mondal**, A. Bhowal, S. Kashyap, R. S. Kshetrimayum, M. Patra, "Performance Analysis of a Mixed RF-FSO Aided Cognitive Radio NOMA Network", Proc. of *International Conference on Microwave, Optical and Communication Engineering (ICMOCE-2023)*, IIT Bhubaneswar, India, May, 2023, pp. 1-4, **Link**.
- [C07.] **S. Mondal**, A. Bhowal, S. Kashyap, R. S. Kshetrimayum, M. Patra, "Outage probability analysis of hard-switching based mixed FSO/IRS-aided RF communication", Proc. of *29th National Conference on Communications (NCC)*, IIT Guwahati, India, Feb, 2023, pp. 1-6, **Link**.
- [C06.] **S. Mondal**, S. D. Roy, and S. Kundu, "Adaptive Energy Harvesting with Relay Selection Schemes in an Ordered NOMA Network", in *IEEE 26th National Conference on Communications (NCC)*, IIT Kharagpur, India, Feb, 2020, pp. 1-6., **Link**.
- [C05.] **S. Mondal**, S. D. Roy, and S. Kundu, "Outage Analysis in Dual Hop Relay Network with an Adaptive Energy Harvesting Scheme", in *IEEE Calcutta Conference (CALCON)*, Kolkata, India, Feb, 2020, pp. 292-296., **Link**.

- [C04.] N. Mishra, **S. Mondal**, S. D. Roy, and S. Kundu, "Cognitive Machine to Machine Communication with Energy Harvesting in IoT networks", in *IEEE 11th International Conference on Communication Systems & Networks (COMSNETS)*, Bangaluru, India, January, 2019, pp. 672–677., **Link**
- [C03.] **S. Mondal**, S. D. Roy, and S. Kundu, "Outage analysis of a multihop cognitive network with energy harvesting from a primary cluster", in *IEEE Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, Bangaluru, India, May, 2017, pp. 81-85., **Link**
- [C02.] S. D. Roy, **S. Mondal**, and S. Kundu, "A new algorithm for admission control of secondary users in CDMA based Cognitive Radio Network", in *International Conference on Computer and Communication Technology (IC CCT)*, Allahabad, India, September, 2010, pp. 35-39., **Link**
- [C01.] S. D. Roy, **S. Mondal**, and S. Kundu, "Performance of joint admission and power control algorithms in cognitive-CDMA network", in *Second International conference on Computing, Communication and Networking Technologies (ICCCNT)*, Karur, India, July, 2010, pp. 1-6., **Link**
- Book Chapters
 1. Sudeshna Sani, Dipra Mitra, and **S. Mondal**, Chapter 22. "Detection of Polarity in the Native-Language Comments of Social Media Networks", Routledge Taylor and francis group, eBook-ISBN-9781003328414 2023.**Link**
 2. I. Bala, **S. Mondal** D. bepari, "Digital twins for industry 4.0 and 5.0", Taylor and francis group, eBook-ISBN-9781003496410.**Link**

WORK EXPERIENCE

- ❑ **Post held:** Assistant Professor in ECE Dept, Bengal College of Engineering & Technology, Durgapur, **Employment period:** August, 2011 - August, 2014
Subject taught: Analog Communication, Digital Electronics, Control System
- ❑ **Post held:** Junior Research Fellow in Central Mechanical Engineering Research Institute, Durgapur, **Employment period:** January, 2010 - August, 2011
Responsibilities: Controller design of "5-Axis, μ -Milling machine", data analysis using Matlab.

AWARDS AND SCHOLASTIC ACHIEVEMENTS

- Gate-2008: Qualified with 90.77 Percentile, All India Rank-2571 in Electronics and Comm. Engg, Reg. No.- EC405223
- Ministry of Human Resource Development (MHRD) Scholarship for M.Tech: 2008-2010
- PhD fellowship from Visvesvaraya PhD scheme under Ministry of Electronics and Information Technology, India : 2015-2020
- Recognized as an Exemplary Reviewer of the IEEE Communication Letters for the year 2022.
- Best Paper Award from "International Conference on ICT Convergence", Oct. 2025, Jeju Island, Korea
- Elevated to IEEE Senior member (93953868): 2024

Patents Filed

1. D. Bepari, B. Acharya, **S. Mondal**, H. K. Gupta, "n-Adjacent Power Allocation to Enhance Capacity in OFDM-based CR Systems", Patent Application No. 202311021007 A, Patent Office Journal No. 20/2023, PP 37115, India, Date of filing of Application: 24/03/2023, Date of Publication: 19/05/2023, **Link**

SHORT TERM COURSES ATTENDED

- Intelligent Reflecting Surface , Organized by: IEEE ComSoc Bangalore, 30th January 2021 to 30th January 2021.
- 5G technologies – Massive MIMO, Millimeter (mm) wave MIMO, NOMA, Full Duplex (FD), OFDM/FBMC and NB-IoT, Organized by: IIT Kanpur, 08th July 2019 to 14th July 2019.

- Bombay Information Theory Seminar 2018, Organized by: IIT Bombay, & TIFR, 11th Jan 2018 to 14th Jan 2018.
- Evolution of Wireless Communication Towards 5G, Organized by: Indian Institute of Technology Kharagpur Under GIAN courses, 07th May 2018 to 17th May 2018.
- Advanced Wireless Networks: From 5G to Cognitive Radio, Organized by: Indian Institute of Engineering Science & Technology, Shibpur Under GIAN courses, 18th Dec 2017 to 22nd Dec 2017.
- Two-week ISTE workshop on Signal and System, Organized by: National Institute of Technology Durgapur, India, Under NMEICT, 02nd Jan 2014 to 12th Jan 2014.

RESEARCH INTERESTS

- Reconfigurable Intelligent Surface, Non-Orthogonal Multiple Access, Backscatter Communication, Free-Space Optical Communication, Vehicular Communication, Integrated Sensing and Communication, Cognitive Radio Networks, Energy Harvesting

PROFESSIONAL ACTIVITIES

- Reviewer (Journals): IEEE Communications Letters (32), IEEE Internet of Things Journal (31), International Journal of Communication Systems (10), IEEE Transactions on Vehicular Technology (8), IEEE Transactions on Communications (6), IEEE Transactions on Wireless Communications (6) **Link**
- Reviewer (Conference): TPC-NCC-2023 ,GCON-2023,Globecom-2025, 2026, ICC-2025

PROFESSIONAL AFFILIATION

- Professional Member, IEEE (93953868)
- Professional Member, Optica – Advancing Optics and Photonics Worldwide (Formerly-OSA) (2051019)
- Life Member, The Institution of Electronics and Telecommunication Engineers (IETE) (M-501865)

PERSONAL INFORMATION

- Permanent Address: 15/S, RD-106, Pursha, Post-Durgapur-7, West Bengal
- Date of Birth: 06th January 1987
- Marital Status: Married
- Category: General
- Languages: English, Bengali and Hindi

REFERENCES

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National Sun Yat-sen University, Kaohsiung, Taiwan
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